PULSE

Master Product, Technical & Operational Specification
Version 3.0 — Consolidated Specification
September 2026
This version consolidates the original Pulse blueprint with all subsequent business-rule, governance, financial, privacy, security, and technical decisions. Where the original blueprint conflicts with later confirmed decisions, the later confirmed decision governs.

1. Executive Summary & Core Intent

Pulse is a digital micro-savings, rotation, community-credit and group-finance platform built around independent Chama containers. A single Pulse account may participate in multiple containers, while every container remains financially and operationally isolated.

Containers support 5–10 members at startup.

Members contribute weekly according to the container's configured contribution rule.

Each container independently controls its rotation/reserve split, loan rules, fines, repayment period and dividend distribution rules.

Financial balances are derived from immutable transaction history rather than directly editable balance fields.

V1 uses manual M-Pesa payment recording and Accountant verification; automatic M-Pesa integration is a future phase.

Pulse provides role-based controls for MEMBER, ACCOUNTANT, CHAIRMAN and SYSTEM_ADMIN.

The platform is also intended as a full-stack engineering, cloud deployment, fintech-architecture and data-analytics project, using the original stack direction: React.js (Vite or Next.js), Tailwind CSS/Lucide, Node.js/Express, MongoDB Atlas/Mongoose and Vercel.

2. Core Architecture & Container Model

2.1 Containers

Minimum starting membership: 5.

Maximum starting membership: 10.

System Admin creates each container.

System Admin gathers member information/input and configures the container rules.

Once a container starts, its rules are locked except during the annual review after 12 complete months.

A member may belong to multiple containers.

Each container has its own membership, contributions, reserve, rotation, loans, dividends, votes, complaints and financial history.

No money, balance, reserve, loan, dividend or transaction may be transferred or mixed between containers.

System-wide account identity is separate from container membership identity.

2.2 Membership Lifecycle

State	Meaning
INVITED	Valid single-use invitation issued by System Admin.
ACTIVE	Member participates normally.
SUSPENDED	Access/participation restricted for an administratively recorded reason.
EXIT NOTICE	Member has submitted a 14-day exit request.
EXITED	Membership ended; historical records remain.
DECEASED / ESTATE REVIEW	Exceptional estate-settlement state.

A container cannot normally operate below 5 members. If an exceptional event causes the active membership to fall below 5, the container enters REVIEW/RESTRICTED status. No new members may be added to a started container. System Admin resolves the exceptional state, including technical closure where necessary.

3. Contributions, Rotation, Reserve & Dividends

3.1 Contribution Rules

Primary/current contribution: KES 1,000 per member per week.

Future containers may configure a minimum contribution of KES 500 per week.

Contribution amount is configurable before a container starts and becomes locked until annual review.

Each container chooses its rotation/reserve-dividend split before starting.

Primary example: KES 1,000 = KES 800 rotation component + KES 200 dividend/reserve component.

Partial contributions are allowed. The system records the amount paid and the outstanding amount.

A weekly contribution is satisfied only after the full configured amount is received and verified.

3.2 Weekly Clock

All user-facing times use East Africa Time (EAT). Internal timestamps are stored in UTC.

Stage	Rule
Monday 00:00	Contribution period opens.
Friday 19:00	Contribution deadline.
After deadline	One-week grace period.
After grace	KES 150 weekly fine applies to an incomplete obligation; repeated weekly fines continue until cleared.

The KES 150 fine applies across contribution amounts. The exact grace/fine parameters may be configured per container before start, subject to the container's adopted rules.

3.3 Rotation

Rotation order is decided by member vote.

Once confirmed and the container starts, the rotation order is fixed.

If a member exits before their turn, their position is removed and the remaining order continues; the system does not automatically rearrange/reassign other positions.

Rotation schedule is generated when the container starts and cannot be changed by System Admin during the active structure.

Rotation payout formula = current active members × current contribution amount × rotation percentage.

Example: 5 members × KES 1,000 × 80% = KES 4,000; 7 members × KES 1,000 × 80% = KES 5,600.

Accountant approval is required before a rotation payout is finalized.

Duplicate rotation payouts are prevented.

Every scheduled recipient receives the full calculated payout even when some members have not paid; the reserve supplies the missing amount and the member's reserve advance is recorded.

3.4 Reserve & Dividend Pool

Current example: 20% of the KES 1,000 contribution is the reserve/dividend component.

10% of the total weekly dividend-pool contribution goes directly to the System/Admin Pool.

90% remains in the member Dividend Pool.

All loan interest and all fines go 100% directly to the Admin Pool.

The Admin Pool belongs entirely to Pulse/System Admin and is not member property.

Member principal savings are protected and are not used to pay operational expenses.

Dividend distribution is normally performed after one complete rotation/cycle, with the distribution schedule fixed before the container starts.

At distribution, the configured distributable member Dividend Pool is shared equally among active members.

4. Reserve Milestone & Group Utilization

Illustrative current reserve configuration: KES 50,000 threshold, with KES 30,000 protected and KES 20,000 available for group use.

The protected reserve is a hard floor for normal reserve use.

The KES 20,000 available portion may be used for a business purchase, group dinner/brunch, cash distribution, reinvestment, group assets/equipment, member lending, or another permitted group purpose.

Members vote on the proposed use.

Approval requires more than 50% of eligible active members. Examples: 10→6, 9→5, 8→5, 7→4, 6→4, 5→3.

The vote deadline is set by System Admin.

If reserve money is lent to a member, the successful vote determines the terms for that particular reserve-lending use; normal loan rules do not automatically override the voted terms.

After a KES 20,000 use, the protected reserve remains KES 30,000 and the reserve rebuilds toward the configured threshold.

If KES 5,000 of a KES 20,000 reserve is currently lent, the system displays total reserve KES 20,000, available cash KES 15,000 and outstanding reserve loans KES 5,000.

If reserve advances a member's missing contribution, repayment returns to the reserve and is tracked as a separate outstanding balance.

5. Loan Engine

5.1 Eligibility & Limits

There is one normal loan product. 'Emergency loan' is informal terminology, not a separate product.

Every member must complete at least one full rotation before becoming loan-eligible.

Self-guaranteed borrowing limit = 50% of the member's current savings.

With one guarantor, the member may access an additional 25% of savings, for a normal maximum of 75% of savings.

Example: KES 10,000 savings → KES 5,000 self-guaranteed + KES 2,500 guarantor-backed = KES 7,500 total.

One guarantor may guarantee only one borrower concurrently.

Borrower's own 50% collateral is locked while the loan is active.

Guarantor savings are not locked when the guarantee is created; if the borrower defaults, the corresponding guarantor savings become locked.

Requests above the configured maximum are automatically rejected.

5.2 Loan Approval Workflow

1. Member submits loan request.
2. System checks container membership, rotation eligibility, savings, outstanding loans, trust/eligibility and configured limits.
3. If a guarantor is required, the borrower selects one eligible guarantor.
4. Guarantor receives only the requested amount/category and explicitly accepts or rejects.
5. If rejected, the borrower may select another eligible guarantor.
6. Accountant approves the loan disbursement.
7. Disbursement is recorded as a financial transaction and the loan becomes active.

5.3 Loan Terms

Current standard interest: 10% flat.

Full 10% remains payable even if the borrower repays early.

Interest is allocated 100% to the Admin Pool.

Standard repayment: 14 days + 7-day grace + KES 150 weekly fine after grace; these parameters are configurable before container start.

Partial loan repayments are allowed.

Partial repayment is allocated in this order: interest/fines first, then principal.

Principal repayments return to the appropriate container funds.

5.4 Loan Statuses

PENDING → APPROVED → DISBURSED → ACTIVE → REPAYMENT DUE → GRACE PERIOD → DEFAULTED → CLEARED

5.5 Default Recovery

1. Calculate the outstanding obligation.

2. Apply the borrower's available eligible savings/entitlement first.
3. If insufficient, use the guarantor's locked savings for the remaining eligible amount.
4. Once the guarantor obligation is satisfied, release the guarantor and restore guarantor eligibility.
5. If borrower and guarantor funds are insufficient, the remaining balance becomes an outstanding group receivable against the borrower.
6. Restrict the borrower from new borrowing until fully cleared.
7. Record every deduction, lock, release and adjustment in the audit/transaction history.

A default does not automatically wipe debt through the reserve. The reserve is not an automatic debt absorber.

6. Trust & Eligibility Tiers

Trust is system-calculated from measurable behavior and is not a numerical public score.
Suggested configurable tiers:

Tier	Indicative basis
NEW	New/limited history; no sufficient repayment history yet.
RELIABLE	Consistent contributions and satisfactory repayment history.
TRUSTED	Longer consistent membership, successful repayments and responsible guarantees.
HIGHLY TRUSTED	Strong sustained contribution, repayment and guarantee history with no recent serious issues.

Positive factors include on-time repayment, early repayment, consistent contributions, prompt fine payment, successful guarantees and membership length. Negative factors include late payment, default, fines, guarantor problems and repeated missed contributions. Trust is recoverable over time.
System calculates the tier. System Admin may override a tier only with a mandatory recorded reason. Trust may affect loan eligibility/limit, guarantor requirements, priority when funds are limited and permitted privileges. Hard financial ceilings remain in force.

7. Roles & Governance

Role	Core authority
MEMBER	Participates in containers, contributes, votes, requests loans, accepts/rejects

	guarantor requests and views permitted records.
ACCOUNTANT	Verifies payments and approves financial transactions, including contributions, withdrawals, reserve advances, loan disbursements, rotation payouts, dividends, Admin Pool withdrawals and exit settlements.
CHAIRMAN	Container leadership role; handles complaints and makes a recommendation/ decision for the matter. No default financial approval authority.
SYSTEM_ADMIN	System-wide administration, container creation/configuration, appointment/ removal of Accountant/Chairman, member administration, technical oversight, final dispute decision and audit oversight.

System Admin is also a participating member.

Roles may vary by container. A person may be Accountant in one container and Member in another.

System Admin cannot approve their own financial transaction; it routes to Accountant.

If System Admin is also Accountant in a container, they still cannot approve their own transaction.

Accountant approval of their own transactions is explicitly permitted by the current business rule.

Role changes are audited and historical actions retain the role held at the time.

8. Complaints & Disputes

Complaints are visible to members of the affected container only.

Complainant may choose named or anonymous submission.

Record includes complainant/anonymous flag, accused/person involved, container, description, date/time, status, Chairman response, System Admin final decision, resolution and audit history.

Workflow: SUBMITTED → UNDER REVIEW → CHAIRMAN RESPONSE → ADMIN DECISION → RESOLVED.

Chairman handles the complaint first; System Admin has final authority.

9. Annual Review

At 12 complete months, the system automatically opens the annual review; System Admin may also open it manually.

Members are notified and may submit rule-change proposals.

System Admin validates proposals before voting and may reject a technically invalid proposal with a mandatory recorded reason.

There is no separate proposal-approval vote before the main decision vote.

System Admin sets the vote deadline.

Only eligible active members vote; suspended/defaulted members are excluded. Members in EXIT NOTICE remain eligible.

One vote per eligible active member regardless of contribution amount.

Votes are anonymous to ordinary members; authorized staff may access individual vote records for audit.

Members may change their vote before the deadline; after the deadline votes are immutable.

Abstention/non-voting is not a 'No' vote.

Approval requires more than 50% of all eligible active members, not merely those who voted.

Failed proposals close without changing the rule and may be resubmitted later.

Approved changes take effect from the next new contribution period, not mid-week.

Existing loans retain the rules under which they were issued.

Existing rotation schedules remain unchanged until the current structure finishes.

Historical records are never rewritten.

10. Membership Exit

Member submits a 14-day exit request through Pulse.

System calculates the earliest exit date and prevents duplicate requests.

Member may cancel the request during the notice period.

During exit notice, normal access remains, but the member does not continue scheduled contributions.

System calculates final settlement; Accountant approves.

Once approved, settlement is frozen. Corrections require a formal audited adjustment.

Exit settlement includes the member's equal share of actual reserve balance at exit, accumulated Dividend Pool share through exit, less outstanding loan, fines, reserve advances and outstanding rotation obligations.

If obligations exceed the settlement, the system shows the amount still owed rather than a negative payout. The member remains outstanding until cleared.

Membership becomes EXITED; records are retained.

Former members may view their historical records but not current information for containers they left.

Pulse account remains active if the person belongs to other containers; the account is not deleted simply because the last membership ends.

11. Death / Estate Handling

Keep this workflow intentionally lightweight. On verified notice, membership enters DECEASED / ESTATE REVIEW; Admin and Accountant are notified; claimant/estate documentation is verified; obligations are deducted first; remaining eligible funds are paid to the verified beneficiary/estate. The deceased member's own locked savings are included. A guarantee they gave another member is not automatically treated as their estate's personal debt. Historical records remain.

12. M-Pesa & Transaction Recording — V1

V1 uses manual M-Pesa recording and verification because Pulse is at the start of its deployment. Automatic M-Pesa integration is a future phase.

Member submits payment details including M-Pesa transaction code, amount, sender phone number, transaction type, period/week and container.

Payment enters Accountant's pending verification queue.

Accountant manually reconciles the submitted code against the relevant bank/M-Pesa record and approves or rejects it.

Rejected payments require a rejection reason/note and remain recorded.

Duplicate M-Pesa transaction codes are rejected globally across Pulse.

Primary and secondary phone numbers may be registered on an account.

Payments from the primary or registered secondary number can be recognized as member payments.

An unregistered third-party sender can be submitted as a third-party payment and requires Accountant manual verification.

Payouts to members are sent to the primary registered phone number.

Members cannot see other members' M-Pesa codes; authorized Admin/Accountant views are controlled by permission.

Future automatic integration should preserve the same transaction/audit model rather than bypassing it.

13. Account, Authentication & Security

Registration is invitation-only through a secure single-use System Admin token.

Invitation expiry: 24 hours.

Email is required.

One primary phone number plus one secondary phone number may be stored.

Passwords are securely hashed and never stored in plain text.

Forgot-password flow is supported; reset tokens are single-use and expire.

Four failed login attempts trigger protection with rate limiting/lockout.

Logout invalidates the active session.

7-day persistent login is disabled by the confirmed rule; sessions must follow the shorter secure session policy configured by the application.

Protected APIs require authentication and backend authorization checks on every request.

Users cannot directly modify database records.

Backend validates identity, container membership, permissions, amount, eligibility, funds and record status for financial operations.

Multi-step financial operations are atomic; failures roll back safely.

Duplicate submission protection is required.

Network failures should trigger status checks before the user resubmits.

14. Privacy & Container Isolation

Individual savings, active loan debt and personal transaction history are private.

Members see aggregate container metrics permitted by the product, such as reserve total, active rotation recipient and pool growth.

Guarantors see only the loan information required for their decision.

Every financial record belongs to exactly one container.

Cross-container financial mixing is technically prohibited.

System Admin has system-wide visibility for administration, but backend authorization still enforces container boundaries for operations.

15. Notifications & Scheduled Jobs

In-app and email notifications are supported in V1; SMS can be added later.

Notifications cover Accountant approvals/rejections, loan approval, guarantor decisions, defaults, voting, rotation and major financial events.

Notification history stores read/unread state; important financial notifications are retained.

Scheduled backend jobs handle weekly cycles, deadlines, grace periods, fines, defaults, rotation reminders, voting deadlines and other configured events.

Failed jobs are logged, surfaced to System Admin, safely retried and protected against duplicate execution.

Admin may manually retry a failed job.

16. Audit & Financial Integrity

Transaction tracking is used; this specification does not require a true accounting double-entry ledger.

Every financial transaction has a permanent unique ID/reference.

Financial records are never physically deleted.

Corrections are made through reversal/adjustment records that preserve the original transaction.

Accountant rejection and reversal require a reason.

Balances are calculated from transaction history.

Zero or negative transactions are rejected except authorized accounting adjustments.

Concurrency protection is required for simultaneous reserve use and other competing financial operations.

Audit records contain actor, role, container, action, timestamp, affected record, old/new value and reason where applicable.

Audit records are immutable; even System Admin cannot delete them.

17. Admin Pool

The Admin Pool is separate from member funds.

10% of weekly Dividend Pool contribution goes to Admin Pool.

100% of loan interest goes to Admin Pool.

100% of fines goes to Admin Pool.

Admin Pool may be used for approved operational/maintenance purposes such as hosting, infrastructure, verification costs and other legitimate Pulse operating needs.

Admin Pool withdrawals require Accountant approval.

Withdrawal record must include amount, purpose, date, approver and resulting balance.

System Admin cannot approve their own Admin Pool withdrawal.

18. Technical Stack

Layer	Confirmed direction
Frontend	React.js with Vite or Next.js; Tailwind CSS; Lucide-React.

Backend	Node.js + Express.js, modular REST API structure.
Authentication	Secure HTTP-only cookie-based JWT approach, subject to the session policy above.
Jobs	Backend scheduled jobs with retry/ idempotency controls.

19. Core Data Model

The original blueprint defines User, Transaction, Loan and Container schemas. The final implementation should expand these into membership-aware and audit-safe structures because one user can belong to multiple containers.

Entity	Purpose
User	Global identity, credentials, primary/ secondary phone, email and account status.
Membership	User-to-container relationship, membership status, role in that container, dates and container- specific state.
Container	Rules, member count, lifecycle, contribution settings, rotation settings, reserve settings, loan settings and governance settings.
Transaction	Immutable financial event with user/ container linkage, type, amount, reference, verification status and audit metadata.
Loan	Borrower/container, principal, flat interest, repayment state, due dates, guarantor relationships, locks and recovery information.
Guarantee	Borrower, guarantor, loan, amount covered, state, lock/release history.
RotationSchedule	Container cycle, ordered members, payout calculation and payout status.
Vote	Proposal, container, eligible population, individual votes, deadline, threshold and final result.

Complaint	Container complaint workflow, participants, responses, decision and audit history.
Notification	Recipient, container, event, message, read state and retention class.
AuditLog	Immutable actor/role/container action history.
Invitation	Single-use registration token, expiry, issuer and consumption status.

20. Core API / Backend Enforcement Requirements

Every protected endpoint authenticates the caller.

Every protected financial endpoint verifies the caller's authorization for the specific container and record.

Server-side rules are authoritative; client-side checks are only UX aids.

Container IDs supplied by clients must never be trusted without membership/role validation.

Financial amounts are represented as exact KES monetary values using integer minor-unit strategy or another precise representation; floating-point arithmetic must not be used for money.

Financial operations must be idempotent where repeated requests could otherwise create duplicate money movement.

All approval endpoints must reject duplicate approval attempts.

All state transitions must validate the current state before changing it.

All money-affecting actions create an audit event and transaction/reference.

21. Important UI / Dashboard Requirements

21.1 Member Dashboard

Personal savings/equity.

Current self-guaranteed loan amount.

Additional guarantor-backed amount.

Maximum borrowing capacity.

Existing outstanding loans and repayment state.

Rotation schedule.

Container reserve aggregate.

Contribution/payment submission.

Loan request and guarantor decisions.

Votes, complaints and notifications.

21.2 Accountant Dashboard

Pending manual M-Pesa verification queue.

Approve/reject deposits with notes.

Financial transaction reconciliation.

Loan disbursement approvals.

Rotation payout approvals.

Dividend distribution approvals.

Reserve advances and withdrawals.

Admin Pool withdrawal approvals.

Exit settlement approvals.

Financial audit history.

21.3 Chairman Dashboard

Container complaints.

Complaint investigation/status workflow.

Response/recommendation submission.

Relevant container governance items.

Chairman does not receive automatic financial approval authority.

21.4 System Admin Dashboard

Create/manage containers.

Configure rules before start.

Manage invitations and memberships.

Appoint/remove Accountant and Chairman.

System analytics and pool health.

Trust-tier management and justified overrides.

Review proposals and annual reviews.

System logs, jobs and audit monitoring.

Container review/restricted/closure management.

22. Container Lifecycle

State	Meaning
DRAFT	Created and configured but not started.
OPEN / FORMING	Members are being invited/confirmed; membership can reach 5–10.
ACTIVE	Rules locked and financial/rotation operations live.
REVIEW / RESTRICTED	Exceptional state such as falling below minimum membership or technical governance issue.
CLOSED	Operations ended after authorized closure and final financial reconciliation; history retained.

23. Non-Negotiable Invariants

1. One user account may have multiple container memberships.
2. One financial transaction belongs to exactly one container.
3. Container money never mixes with another container.
4. No new member joins a started container.
5. Rules are locked during normal operation and become reviewable after 12 complete months.
6. All interest and fines go 100% to Admin Pool.
7. 10% of the configured Dividend Pool contribution goes to Admin Pool; 90% remains member Dividend Pool.
8. Protected reserve is not reduced below its configured floor by normal reserve use.
9. Rotation recipient receives the full calculated payout.
10. Loan eligibility begins only after one complete rotation.
11. Normal loan ceiling is 50% savings self-guaranteed plus 25% additional with one guarantor.
12. Accountant approves financial disbursements and settlements.

13. System Admin cannot approve their own financial transaction.
14. Financial records are never physically deleted.
15. Audit records are immutable.
16. V1 M-Pesa verification is manual.

24. Superseded Original-Blueprint Items

The original uploaded blueprint remains the source baseline, but several of its rules are superseded by later confirmed decisions. The original document described a 10-member fixed model, a 7–14 day emergency-loan framing, 2 guarantors, a 2.5× savings cap, KES 100 fines, 85/15 interest-and-fine revenue allocation, consultative leadership for the KES 20,000 reserve use, and 7-day persistent sessions. Those items must not be implemented where they conflict with this consolidated specification. The original blueprint also called its ledger 'double-entry'; the confirmed model is transaction tracking rather than a true double-entry accounting system. The uploaded blueprint itself contains the earlier versions of these rules. filecite turn2file0 filecite turn3file4
The original blueprint already identified manual verification as part of its transaction model, including an Accountant pending queue and rejection notes; this consolidated version makes manual recording explicitly the V1 approach and preserves the reconciliation/audit concept. filecite turn3file5

25. Implementation Principle

Pulse should be implemented from this consolidated specification as the authoritative business-rule document. Where a screen, API, database model or scheduled job conflicts with an invariant in this specification, the invariant takes precedence. Any future change must be recorded as a versioned rule change rather than silently modifying historical behavior.

End of Pulse Master Product, Technical & Operational Specification — Version 3.0